

103 - Introduction to computer

Model paper – 1

Marks-50

Q – 1 (A) Attempt any Five

5

1. The High-speed memory between the main memory and CPU is called.
A. Register Memory B. Storage Memory
C. Cache Memory D. Virtual Memory
2. SRAM Stands for
A. Static Random Access Memory
B. Static Read Access Memory
C. Stable Random Access Memory
D. Static Random Arithmetic Memory
3. Computer with highest speed and capacity is:
A. Mini Computer B. Super Computer
C. Micro Computer D. Mainframe Computer
4. Which of the following can be output by a computer?
A. Text B. Graphics
C. Voice D. All of these
5. Which of the following is not a web Browser?
A. Mozilla Firefox B. Google Chrome
B. windows XP D. Safari
6. Which of the following is NOT considered a pointing device?
A. Mouse B. Trackball
C. Joystick D. Keyboard

7. Which of the following can be output by a computer?

A. Text

B. Graphics

B. Voice

D. All of these

Q – 1 (B) Answer The following in Short (Any 4)

8

1. State the difference between Volatile Memory & Non-Volatile Memory?
2. Explain URL with an example.
3. Name some application where you use touch screen.
4. Give full form of OLED, TFT, LCD and CRT.
5. What is the use of EPROM?

Q – 2 (A) Answer the following in detail.

- (1) Explain Characteristics of the computer.

8

OR

- (1) Draw a Block diagram of computer and explain its functional units.

8

Q – 2 (B) Answer the following in detail.

- (1) Explain motherboard in detail.

5

Q – 3 Do as Directed (Attempt any 2)

12

Q – 3 (A) Attempt following.

- (1) Subtract: $(10111)_2$ from $(11011)_2$
- (2) $(356)_{10} = (?)_2$
- (3) $(9AB)_{16} = (?)_{10}$

Q – 3 (B) Attempt following.

- (1) $(10001100.110)_2 = (?)_{10}$
- (2) Add $(11101.11)_2$ to $(11011.10)_2$
- (3) $(1715)_{10} = (?)_2$

Q – 3 (C) Answer in Brief.

- (1) What is RFID? Explain any one application in which RFID is used in detail.

Q – 4 Answer in Brief. (any 2)

12

- (1) What is mechanical mouse? How it differs from optical mouse.
- (2) what do you mean by cloud? Explain various application of cloud.
- (3) Differentiate Inkjet and Laser Printer.



Internal Examination : B.C.A. (Regular)

Subject: Introduction To Computer

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Seat No:

Q-1 [A] MCQS (Any 5)

[5 Marks]

- Which of the following is not used as a portable computer?
 - Desktop computer
 - Tablet
 - Laptop
 - Notebook computer
- RAM is :
 - Volatile
 - Non-Volatile
 - Temporary
 - None
- There are _____ number of characters represented ASCII
 - 128
 - 256
 - 60
 - 23
- Light Pen is used for:
 - Printing
 - Drawing On Screen
 - Store Data
 - None of these
- WWW Stands for:
 - World Wide Web
 - Web Wide World
 - Wide Web World
 - All of these
- LED Stands for:
 - Light Emitting Diode
 - Light Crystal Diode
 - Laser Crystal Diode
 - All of these
- What is the Full form of NIC?
 - Network Interface Card
 - None of these
 - Net Interconnect Card
 - Network Card

Q-1 [B] Attempt Any Four out of Five

[8 Marks]

- $(1234)_{10} = (?)_2$
- $(2FB)_{16} = (?)_{10}$
- Perform : $(11110.01)_2 + (01011.11)_2$
- Subtract $(0110110.01)_2$ From $(1000101.01)_2$
- $(10011010.01)_2 = (?)_{10}$

Q-2 [A] Answer the following in Details

[8 Marks]

- Explain Block diagram of computer and write the functional units with suitable diagram?
OR
- Explain ROM ? And Explain its type in Details?

Q-2 [B] Answer the following in Details

[8 Marks]

- What is Printer? Write any one Printer in details with pros and cons?

Q-3 Answer the following in Details (Attempt any Two out of Three)

[12 Marks]

1. Difference Between Super Computer and Mainframe Computer
2. What do you mean by Cache Memory and Virtual Memory in details? Draw a neat diagram of Cache memory?
3. Explain Internet Through Cable in details with suitable Diagram?

Q-4 Answer in Brief (Attempt any Two out of Three)

[12 Marks]

1. What is Optical Disk? Explain Its type in details? What are the Advantage and Disadvantage of Optical Disk?
2. What is RFID? Explain Application of RFID?
3. List out Internet Services? Explain Cable Modem Services in Details with suitable Diagram?

Unit 1 – Introduction of Computer

1. A computer is an electronic device that:
 - a) Accepts data
 - b) Processes data
 - c) Produces output
 - d) All of the above**Answer: d) All of the above**
2. The speed of a computer is measured in:
 - a) Seconds
 - b) Minutes
 - c) Milliseconds, microseconds, nanoseconds
 - d) Hours**Answer: c) Milliseconds, microseconds, nanoseconds**
3. Errors in computers mostly occur due to:
 - a) Machine fault
 - b) Human errors
 - c) Power supply
 - d) Network issues**Answer: b) Human errors**
4. The term **GIGO** means:
 - a) Great Input Good Output
 - b) Garbage In Garbage Out
 - c) General Information General Output
 - d) Good Information Great Output**Answer: b) Garbage In Garbage Out**
5. The ability of a computer to perform repetitive tasks without boredom is called:
 - a) Versatility
 - b) Diligence
 - c) Accuracy
 - d) Speed**Answer: b) Diligence**
6. Which of the following is a limitation of computers?
 - a) No IQ
 - b) No feelings
 - c) Cannot work on its own
 - d) All of the above**Answer: d) All of the above**
7. Which type of computer is the most powerful?
 - a) Microcomputer
 - b) Supercomputer
 - c) Minicomputer
 - d) Mainframe**Answer: b) Supercomputer**
8. PARAM 10000 is an example of:
 - a) Microcomputer
 - b) Supercomputer
 - c) Mainframe computer

d) Laptop

Answer: b) Supercomputer

9. Computers used for bulk data handling and airline reservations are:

a) Microcomputers

b) Minicomputers

c) Mainframe computers

d) Supercomputers

Answer: c) Mainframe computers

10. A small, inexpensive general-purpose computer with a microprocessor as its CPU is:

a) Supercomputer

b) Mainframe

c) Minicomputer

d) Microcomputer

Answer: d) Microcomputer

11. Which of the following is a portable computer?

a) Desktop

b) Laptop

c) Supercomputer

d) Mainframe

Answer: b) Laptop

12. A group of 8 bits is called:

a) Nibble

b) Byte

c) Word

d) Register

Answer: b) Byte

13. The main components of the CPU are:

a) ALU, CU, Registers

b) RAM, ROM, Hard disk

c) Input, Output, Storage

d) None of the above

Answer: a) ALU, CU, Registers

14. Which memory is volatile?

a) RAM

b) ROM

c) Hard Disk

d) DVD

Answer: a) RAM

15. Which of the following is **primary storage**?

a) RAM

b) Hard Disk

c) CD-ROM

d) DVD

Answer: a) RAM

Unit 2 – Basic Computer Architecture

1. Computer system architecture shows the relationship between:

a) Hardware, Software, User

- b) Input, Output, Storage
- c) CPU, Memory, Registers
- d) None of the above

Answer: a) Hardware, Software, User

2. Which of the following is an example of **system software**?
- a) MS Word
 - b) Hotel Management System
 - c) Compiler
 - d) Tally ERP

Answer: c) Compiler

3. Which of the following is **application software**?
- a) Windows 10
 - b) Antivirus
 - c) Payroll System
 - d) BIOS

Answer: c) Payroll System

4. The main circuit board of a computer is called:
- a) Processor
 - b) Motherboard
 - c) Hard disk
 - d) Cache

Answer: b) Motherboard

5. Which of the following is **not a component of a motherboard**?
- a) Expansion slots
 - b) SMPS
 - c) System Clock
 - d) NIC

Answer: b) SMPS

6. Which bus carries information about **where data is stored in memory**?
- a) Data Bus
 - b) Address Bus
 - c) Control Bus
 - d) System Bus

Answer: b) Address Bus

7. The function of **Control Bus** is:
- a) Carry data only
 - b) Carry address only
 - c) Carry control and timing signals
 - d) None of the above

Answer: c) Carry control and timing signals

8. Which processor is mainly designed for **graphics rendering**?
- a) CPU
 - b) GPU
 - c) ALU
 - d) CU

Answer: b) GPU

9. CPU is more suitable for:
- a) Parallel instruction processing
 - b) Serial instruction processing
 - c) Only arithmetic operations

d) Data storage

Answer: b) Serial instruction processing

10. The speed of a CPU is measured in:

a) Bytes

b) Hertz (Hz)

c) Megabytes

d) Joules

Answer: b) Hertz (Hz)

11. Memory that loses data when power is switched off is called:

a) Non-volatile memory

b) Secondary memory

c) Volatile memory

d) Optical memory

Answer: c) Volatile memory

12. Which of the following is **non-volatile memory**?

a) RAM

b) Cache

c) ROM

d) Register

Answer: c) ROM

13. Which memory is faster but more expensive?

a) RAM

b) Cache Memory

c) Hard Disk

d) DVD

Answer: b) Cache Memory

14. The process of transferring data between RAM and Hard disk during program execution is called:

a) Buffering

b) Paging

c) Interrupt

d) Booting

Answer: b) Paging

15. Which of the following storage devices is **optical**?

a) Hard disk

b) DVD

c) USB Pen Drive

d) SSD

Answer: b) DVD

Unit 3 – Number System

1. The base of **binary number system** is:

a) 8

b) 2

c) 10

d) 16

Answer: b) 2

2. The base of **octal number system** is:

- a) 2
- b) 8
- c) 10
- d) 16

Answer: b) 8

3. The base of **hexadecimal number system** is:

- a) 2
- b) 8
- c) 10
- d) 16

Answer: d) 16

4. The base of **decimal number system** is:

- a) 2
- b) 8
- c) 10
- d) 16

Answer: c) 10

5. Which digits are used in hexadecimal number system?

- a) 0–7
- b) 0–9, A–F
- c) 0–15
- d) 0–9 only

Answer: b) 0–9, A–F

6. A single binary digit is called:

- a) Byte
- b) Word
- c) Bit
- d) Nibble

Answer: c) Bit

7. A group of **4 bits** is called:

- a) Byte
- b) Word
- c) Nibble
- d) Register

Answer: c) Nibble

8. A group of **8 bits** is called:

- a) Byte
- b) Word
- c) Register
- d) Octet

Answer: a) Byte

9. The largest single digit in octal system is:

- a) 7
- b) 8
- c) 9
- d) F

Answer: a) 7

10. The largest single digit in hexadecimal system is:

- a) 9

- b) E
- c) F
- d) 10

Answer: c) F

11. Which number system is most suited for computers?

- a) Decimal
- b) Binary
- c) Hexadecimal
- d) Octal

Answer: b) Binary

12. $(10101)_2 = (?)_{10}$

- a) 20
- b) 21
- c) 22
- d) 23

Answer: b) 21

13. $(1111)_2 = (?)_{10}$

- a) 12
- b) 13
- c) 14
- d) 15

Answer: d) 15

14. $(999)_{10} = (?)_2$

- a) 1111100011
- b) 1111100111
- c) 1111101000
- d) 1111100110

Answer: b) 1111100111

15. ASCII uses how many bits to represent a character?

- a) 8 bits
- b) 16 bits
- c) 7 bits
- d) 6 bits

Answer: c) 7 bits

Unit 4 – Input & Output Devices

1. An input device is used to:

- a) Display output
- b) Provide data to the computer
- c) Store information permanently
- d) Control programs

Answer: b) Provide data to the computer

2. A **pointing device** controls:

- a) CPU speed
- b) Cursor movement
- c) Memory capacity
- d) Hard disk access

Answer: b) Cursor movement

3. A **mechanical mouse** works on:
- a) LED light reflection
 - b) Laser beam
 - c) Rubber ball movement
 - d) Capacitive sensor
- Answer: c) Rubber ball movement**
4. An **optical mouse** uses:
- a) Ball mechanism
 - b) Light and sensor
 - c) Magnetic field
 - d) Track wheel
- Answer: b) Light and sensor**
5. A **trackball** differs from a mouse because:
- a) It uses keyboard
 - b) Ball is stationary, user rotates it
 - c) It is input & output device both
 - d) It works without cursor
- Answer: b) Ball is stationary, user rotates it**
6. Which device is widely used for playing games?
- a) Mouse
 - b) Joystick
 - c) Light pen
 - d) Scanner
- Answer: b) Joystick**
7. A **light pen** is used to:
- a) Print documents
 - b) Select objects directly on screen
 - c) Scan documents
 - d) Store data
- Answer: b) Select objects directly on screen**
8. The working of a **keyboard** involves:
- a) System clock
 - b) Keyboard buffer
 - c) Hard disk
 - d) BIOS only
- Answer: b) Keyboard buffer**
9. **FASTag** system is based on:
- a) Barcode
 - b) RFID
 - c) OCR
 - d) MICR
- Answer: b) RFID**
10. Which device converts paper documents into digital form?
- a) Joystick
 - b) Scanner
 - c) Printer
 - d) Plotter
- Answer: b) Scanner**
11. Barcode reader uses:
- a) Magnetic ink

- b) Optical beams
- c) Laser engraving
- d) RFID tags

Answer: b) Optical beams

12. A **web camera** does not store images because:

- a) It has no lens
- b) It directly transfers images to computer
- c) It is only for video calls
- d) It uses battery

Answer: b) It directly transfers images to computer

13. Which of the following is an **impact printer**?

- a) Inkjet printer
- b) Laser printer
- c) Dot matrix printer
- d) Thermal printer

Answer: c) Dot matrix printer

14. A **non-impact printer** works by:

- a) Hammering inked ribbon
- b) Striking pins on paper
- c) Using laser or ink spray
- d) Punching holes

Answer: c) Using laser or ink spray

15. Which monitor technology provides the **highest contrast and wide viewing angles**?

- a) LCD
- b) LED
- c) TFT
- d) OLED

Answer: d) OLED

Unit 1 – Introduction

1. Define computer.
2. Justify that computer is a versatile device.
3. Differentiate between laptop and desktop.
4. List devices involved by the Input Unit.
5. What is the functionality of the input unit?
6. List devices involved by the Output Unit.
7. What is the functionality of the output unit?
8. What is the work of the memory unit/storage unit?
9. What are the components of CPU?
10. List out the functional units of the computer.

Unit 2 – Basic Computer Architecture

1. How does Computer memory work?
2. What is access time?
3. What are the basic types of memory?
4. Differentiate between primary and secondary memory.
5. What is fixed length memory? Explain.
6. What is variable length memory? Explain.
7. Give full form of: SRAM, DRAM, DDR-DRAM, FCRAM, FPM-DRAM, PROM, EPROM, EEPROM, UVEPROM.
8. What is Virtual Memory? Explain in short.
9. What are the different type of hard disk?
10. What are the different types of optical disks?

Unit 3 – Number System

1. What is ASCII code? Give its full form.
2. What is binary number system? Also write its base and digits.
3. What is an octal number system? Also write its base and digits.
4. What is hexadecimal number system? Also write its base and digits.
5. What is ANSI code? Give its full form.
6. What is the significance of ANSI Character Code?
7. What is Number system? List out various types of number system.
8. What is the importance of radix in number system? Give radix of decimal and binary.
9. What do you mean by character codes? How many bits are used to represent ASCII?
10. Decimal to Binary conversions (e.g., $999_{10} = ?_2$, $1234_{10} = ?_2$).
11. Binary to Decimal conversions (e.g., $101110_2 = ?_{10}$, $1111_2 = ?_{10}$).

Unit 4 – Input & Output Devices

1. What is pointing device?
2. Differentiate optical mouse and mechanical mouse.
3. What is RFID?
4. Differentiate input and output device.
5. Write full form of LED, LCD, TFT and OLED.
6. Describe optical scanner.
7. Differentiate impact and non-impact printer.
8. List various pointing devices.
9. Differentiate inkjet printer and laser printer.
10. What are the advantages of dot matrix printer?
11. What is use of barcode reader?
12. What are the limitation of laser printer?
13. Differentiate dot matrix and inkjet printer.
14. What is the role of Input device in computer?
15. What is the role of Output device in computer?
16. Explain types of printers according to their printing style.
17. Explain types of printers according to technology used at manufacturing.
18. What is FASTag?
19. Explain bar code reader.
20. Define term: trackball.

Unit 5 – Internet Concepts & Cloud

1. What is web browser? List out the name of Web browsers.
2. What is difference between Wi-Fi and Hotspot?
3. What is the use of Bluetooth?
4. List out Advantages and Disadvantages of Internet.
5. Explain use of Modem.
6. Explain URL with an example.
7. What is cloud?
8. Explain the concept of bookmark in Web browser.
9. Explain the uses of router.
10. What are the uses of Fire-Stick?

Unit – 1: Introduction

1. Explain in detail the characteristics of a computer.
2. Write a note on applications of computer.
3. Write a note on types of computer.
4. Compare mini computer and micro computer.
5. Write a short note on portable computers.
6. Draw a block diagram of Computer and explain its functional units.

Unit – 2: Basic Computer Architecture

1. Explain the Computer System Architecture.
2. Explain motherboard in detail.
3. Explain primary or main memory organization in detail.
4. Explain RAM in detail.
5. Explain ROM in detail.
6. Explain various addressing modes of computer memory in detail.
7. What is Cache Memory? Write a short note on that.
8. Explain hard disk in detail.
9. Explain optical disk in detail.

Unit – 3: Number System (Examples only)

1. Convert $(68)_{10}$ to Binary
2. Convert $(125)_{10}$ to Binary
3. Convert $(124.24)_{10}$ to Binary
4. Convert $(120.25)_{10}$ to Binary
5. Convert $(10101)_2$ to Decimal
6. Convert $(101101.001)_2$ to Decimal
7. Convert $(110001.101)_2$ to Decimal
8. Add $(10001)_2$ to $(11101)_2$
9. Add $(1100111)_2$ to $(110011)_2$
10. Add $(11101.11)_2$ to $(11011.10)_2$
11. Subtract $(1101)_2$ from $(11101)_2$
12. Subtract $(1001)_2$ from $(10110)_2$
13. Subtract $(110.0011)_2$ from $(111.0100)_2$

Unit – 4: Input & Output Devices

1. Explain different pointing devices in detail.
2. Explain working of keyboard with neat diagram.
3. Explain the concept of RFID and its application in FASTag.
4. Explain working of Optical Scanner with diagram.
5. Explain Bar Code Reader in detail.
6. Explain Web Camera in detail.
7. Explain different types of monitors (LED, LCD, TFT, OLED, Touch screen) with comparison.

8. Explain types of printers with examples.
9. Explain Impact and Non-Impact printers with comparison.
10. Write a note on 3D printers.

Unit – 5: Internet & Web Technologies

1. What is Internet? Explain different types of Internet services with examples.
2. What do you mean by Cloud? Explain its various applications. Discuss the **advantages and disadvantages** of Cloud.
3. Explain the **concept of online data backup** with a suitable example.
4. Explain the **comparison between various Internet connections** like Hotspot, Wi-Fi, and Cable.
5. What do you mean by E-mail? Explain **TO, SUBJECT, CC, BCC and Attachment** with a neat diagram and suitable example.
6. What do you mean by Web Browser? List out the names of Web browsers and discuss the **various terminologies of web browsers**.